

ScottDomains — Symbol Map

Gunter & Scott, “Semantic Domains” (HTCS Vol. B, 1990), pp. 633–674 — reading the paper and reading/writing the Lean.

Table A — PDF garbling → Unicode (for reading the paper)

The paper’s PDF uses 1990-era Type-3 bitmap fonts; `pdftotext` maps many glyphs to unrelated ASCII and silently drops the `fi`/`fl`/`ff`/`ffi` ligatures. Several distinct operators all collapse to a blank gap — context (sets vs. domains) disambiguates. Snippets below are quoted verbatim from the extraction.

Extraction	Correct	Meaning / example (extraction → intended)
!	→	continuous-map / function-space arrow: $f : D \rightarrow f : D \rightarrow D$
!	ω	the ordinal ω (SAME glyph as the arrow): <code>ordinal !, cpo ! > → ω^T</code>
v	$\sqsubseteq$	the partial order (reflexive relation): <code>binary relation v, x v y</code>
w	$\supseteq$	the reverse order: <code>x w y → x $\supseteq$ y</code>
F	$\sqcup$	directed least upper bound (join): <code>x v F M → x $\sqsubseteq$ $\sqcup$ M</code>
2	$\in$	set membership: <code>x 2 D → x $\in$ D</code>
8	$\forall$	universal quantifier: <code>8x 2 u → $\forall x \in u$</code>
9	$\exists$	existential quantifier: <code>9y 2 E → $\exists y \in E$</code>
>	$\top$	top element: <code>> and ? → $\top$ and $\perp$</code>
?	$\perp$	bottom / least element: <code>?D → $\perp_D$, ? v > → $\perp \sqsubseteq \top$</code>
(blank gap)	$\subseteq$	subset (of sets): <code>M D is directed → $M \subseteq D$; u N</code>
(blank gap)	$\times$	Cartesian product (of domains): <code>D E → D $\times$ E; fst : D E ! D</code>
(blank gap)	$\otimes$	smash / tensor product: <code>smash : D E ! D E → smash : D $\times$ E → D $\otimes$ E; f g → f $\otimes$ g</code>
(blank gap)	$\circ$	function composition: <code>g = f unsmash → g = f $\circ$ unsmash; smash (f g)</code>
(blank gap)	$\cdot$	multiplication / application dot: <code>x x → x $\cdot$ x</code>
7!	$\mapsto$	maps-to: <code>x 7! x x → x $\mapsto$ x $\cdot$ x</code>
(vanished)	λ	Church’s binder drops entirely: <code>x : x x → $\lambda x. x \cdot x$; -notation → λ-notation</code>
line-lead =	$\cong$	isomorphism ($\cong$; the tilde is dropped, leaving a bare = at line start): <code>D <newline> = D ! D → D $\cong$ (D → D)</code>
{	–	en-dash in number/page ranges: <code>633{674 → 633–674</code>
h ...i	$\langle \dots \rangle$	tuple / pairing angle brackets: <code>hf1, : : :, fni → $\langle f_1, \dots, f_n \rangle$</code>
: : :	...	ellipsis (also the dotted leaders in the table of contents): <code>(D1; : : :; Dn) → (D₁, ..., D_n)</code>
fi drop	fi	<code>de nition</code> →definition, <code>nite</code> →finite, <code>xed</code> →fixed, <code>bi nite</code> →bifinite, <code>nitary</code> →finitary, <code>satis</code> <code>es</code> →satisfies, <code>signi cant</code> →significant, <code>rst</code> →first, <code>nd</code> →find
fl drop	fl	<code>re exive</code> →reflexive, <code>brie y</code> →briefly
ff drop	ff	<code>di erent</code> →different, <code>e cts</code> →effects, <code>e ctively</code> →effectively
ffi drop	ffi	<code>diculty</code> →difficulty, <code>dicult</code> →difficult

Table B — Symbol ↔ Unicode ↔ Lean input ↔ Lean display (for the Lean)

Lean input = the `\`-sequence typed in the VS Code `lean4` extension (equivalently `lean4-mode` in Emacs), from the extension’s `abbreviations.json`. The primary canonical abbreviation is listed first; common aliases follow. “Lean shows” is the glyph the editor substitutes — identical to the Symbol column, rendered here through `unicode-math`. Every sequence below was checked against the installed abbreviation table; none are unverified.

Symbol	Unicode (code point + name)	Lean input	Lean shows
$\sqsubseteq$	U+2291 SQUARE IMAGE OF OR EQUAL TO	<code>\sqsubseteq</code> (<code>\squb</code>)	$\sqsubseteq$
$\supseteq$	U+2292 SQUARE ORIGINAL OF OR EQUAL TO	<code>\sqsupseteq</code> (<code>\squp</code>)	$\supseteq$
$\sqcup$	U+2294 SQUARE CUP	<code>\sqcup</code> (<code>\lub</code> , <code>\join</code>)	$\sqcup$
$\sqcap$	U+2293 SQUARE CAP	<code>\sqcap</code> (<code>\glb</code> , <code>\meet</code>)	$\sqcap$
$\sqcup$	U+2A06 N-ARY SQUARE UNION OPERATOR	<code>\bigsqcup</code> (<code>\Sqcup</code> , <code>\Sup</code> , <code>\supr</code>)	$\sqcup$
$\sqcap$	U+2A05 N-ARY SQUARE INTERSECTION OP.	<code>\bigsqcap</code> (<code>\Sqcap</code> , <code>\Inf</code> , <code>\infi</code>)	$\sqcap$
$\perp$	U+22A5 UP TACK	<code>\bot</code>	$\perp$
$\top$	U+22A4 DOWN TACK	<code>\top</code>	$\top$
$\ll$	U+226A MUCH LESS-THAN (way-below)	<code>\ll</code>	$\ll$
$\rightarrow$	U+2192 RIGHTWARDS ARROW	<code>\to</code> (<code>\r</code> , <code>\-></code> , <code>\rightarrow</code>)	$\rightarrow$

Symbol	Unicode (code point + name)	Lean input	Lean shows
$\mapsto$	U+21A6 RIGHTWARDS ARROW FROM BAR	<code>\mapsto (\mapsto)</code>	$\mapsto$
$\circ$	U+2218 RING OPERATOR (composition)	<code>\circ (\circ, \text{comp})</code>	$\circ$
$\times$	U+00D7 MULTIPLICATION SIGN	<code>\times (\times)</code>	$\times$
$\otimes$	U+2297 CIRCLED TIMES (smash / tensor)	<code>\otimes (\otimes)</code>	$\otimes$
$\oplus$	U+2295 CIRCLED PLUS (sum)	<code>\oplus (\oplus)</code>	$\oplus$
$\cong$	U+2245 APPROXIMATELY EQUAL TO (iso)	<code>\cong (\cong, \sim=)</code>	$\cong$
$\simeq$	U+2243 ASYMPTOTICALLY EQUAL TO (equiv)	<code>\simeq (\simeq)</code>	$\simeq$
∞	U+221E INFINITY	<code>\infty</code>	∞
ω	U+03C9 GREEK SMALL LETTER OMEGA	<code>\omega (\omega)</code>	ω
β	U+03B2 GREEK SMALL LETTER BETA	<code>\beta (\beta)</code>	β
λ	U+03BB GREEK SMALL LETTER LAMDA	<code>\lambda (\lambda, \text{fun})</code>	λ
α	U+03B1 GREEK SMALL LETTER ALPHA	<code>\alpha (\alpha)</code>	α
Γ	U+0393 GREEK CAPITAL LETTER GAMMA	<code>\Gamma (\Gamma)</code>	Γ
Π	U+03A0 GREEK CAPITAL LETTER PI	<code>\Pi (\Pi)</code>	Π
Σ	U+03A3 GREEK CAPITAL LETTER SIGMA	<code>\Sigma (\Sigma)</code>	Σ
$\in$	U+2208 ELEMENT OF	<code>\in (\in, \text{mem})</code>	$\in$
$\notin$	U+2209 NOT AN ELEMENT OF	<code>\notin (\notin)</code>	$\notin$
$\forall$	U+2200 FOR ALL	<code>\forall (\forall, \text{all})</code>	$\forall$
$\exists$	U+2203 THERE EXISTS	<code>\exists (\exists)</code>	$\exists$
$\subseteq$	U+2286 SUBSET OF OR EQUAL TO	<code>\subseteq (\subseteq, \text{ss}, \text{sub})</code>	$\subseteq$
$\subset$	U+2282 SUBSET OF	<code>\subset (\subset) [\text{subset} \Rightarrow \subseteq]</code>	$\subset$
$\wedge$	U+2227 LOGICAL AND	<code>\wedge (\wedge)</code>	$\wedge$
$\vee$	U+2228 LOGICAL OR	<code>\vee (\vee)</code>	$\vee$
$\neg$	U+00AC NOT SIGN	<code>\neg (\neg, \text{neg}, \text{lnot})</code>	$\neg$
$\leq$	U+2264 LESS-THAN OR EQUAL TO	<code>\leq (\leq)</code>	$\leq$
$\geq$	U+2265 GREATER-THAN OR EQUAL TO	<code>\geq (\geq)</code>	$\geq$
$\rightsquigarrow$	U+219D RIGHTWARDS WAVE ARROW	<code>\rightsquigarrow (\rightsquigarrow)</code>	$\rightsquigarrow$
$\mathbb{1}$	U+1D7D9 MATH DOUBLE-STRUCK DIGIT ONE	<code>\mathbb{1}</code>	$\mathbb{1}$

Sources: extraction via pdftotext on ScottDomains/papers/Gunter Scott 1990.pdf; Lean input from the lean4 abbreviation table (abbreviations.json).